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Matrix Acid Stimulation - A Review of the State-Of-The-Art

Phil Rae, SPE, and Gino di Lullo, SPE, BJ Services Company

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Abstract

Acid stimulation of oil and gas reservoirs, with a view to increasing well productivity, has been applied since the late-19th century. Initially applied in carbonate reservoirs, the technique was extended to more complex mineralogies, over a number of years. However, it's fair to say that acid stimulation of wells is the exception rather than the rule. This probably stems from the complex, heterogeneous nature of formation minerals and the unpredictability of their response to conventional oilfield acid formulations. With inappropriate acid designs, or poor job procedures, even the best candidate wells can be damaged, sometimes irreversibly.

This paper discusses the current state-of-the art in matrix acidising and makes the case for the wider implementation of acidising, as a cost-effective method for production enhancement. It reviews the many rules used today in the design of acid treatments and how these rules have evolved with improvements in our understanding of the interactions between acids, formation constituents and well tubulars. The paper also reviews the rationale behind the use of additives such as corrosion inhibitors, iron control agents, clay control additives, surfactants, solvents, anti-sludges and diverting agents, etc. and makes general recommendations on appropriate loadings, where applicable.

Finally, the latest developments in acidising are considered, including the use of novel acid systems, to overcome many of the problems inherent in earlier formulations. Innovative equipment design, coupled with real-time monitoring capabilities, improved placement techniques and environmentally-friendly materials, are helping to transform acidising into a valuable asset in the quest for optimum performance from every oil and gas well. The paper references many key publications and provides the engineer with an up-

to-date overview of the state-of-the-art in this very important discipline.

Introduction

Matrix acid stimulation is viewed by many as a risky enterprise and one that should be undertaken only as a last resort. Yet, this relatively simple technique certainly represents one of the most cost-effective methods to enhance well productivity and improve hydrocarbon recovery. When properly applied, it is also an effective way to reduce the environmental impact of our industry, ensuring that reservoir drainage is efficient by optimising productive capacity from previously damaged wells.

The science of acidising has its origins over 100 years ago when Herman Frasch of Standard Oil patented the use of hydrochloric acid to stimulate carbonate formations.⁽¹⁾ Simultaneously, one of his colleagues patented the use of sulphuric acid for the same purpose⁽²⁾. Apart from a brief flurry of activity, resulting from the original idea, neither technique was applied on a widespread basis, during the next thirty years. Then, in the early 1930's several serendipitous events occurred. The Dow Chemical Company developed an effective acid corrosion inhibitor for mineral acids and was asked to provide it for an acid treatment being performed by the Pure Oil Company on one of their wells in Michigan. The effect of the treatment on production was positive, if not spectacular, but it provided the impetus to perform further treatments. Some of these later treatments produced excellent results and news of the technique quickly spread, spawning a whole host of small companies, eager to participate in this new business.⁽³⁾

In the same decade, attempts were made to improve production from sandstone reservoirs by injecting mixtures of hydrochloric and hydrofluoric acids. These early treatments were not particularly successful, however, and this relegated these HCl/HF mixtures to only occasional use in those wells that were deemed to have suffered drilling mud damage. It was not until the 1960's that treatments containing hydrofluoric acid again saw widespread use in well remediation. This was largely due to the publication of studies on the chemical interactions of HF with typical sandstone formation minerals, along with guidelines for treatment optimisation. This work eliminated much of the mystery surrounding the use of HF and put its use in practical terms that petroleum engineers could understand. However, in spite of this, acidising sandstone formations remained a hit-or-miss enterprise. It was fabulously successful in some areas, totally

disastrous in others. Indeed, throughout the 1970's and 80's, most production engineers and service company personnel could recite details of some catastrophic acid job that they had been involved with.

These failures tended to overshadow the successful treatments, not because of greater numbers but because of the "fallout" surrounding such events, in the oilfield. As a consequence, in many areas, acidising activity all but ceased. Yet, of all remedial treatments, small matrix acid jobs, whether they are performed in sandstone or carbonate reservoirs, can, potentially, represent the best return-on-investment due to their relative simplicity and minimal equipment requirements. There are many good examples, in numerous countries, where no acid treatments were carried out for many years in sandstone reservoirs, due to poor or even disastrous results. Bolivia, Brunei, Malaysia and Nigeria are good examples. Recently, new acid systems have been used to restore well productivity so successfully that sandstone acidising have become a routine operation for many producing fields in those countries.

Huge numbers of wells around the world are producing sub-optimally due to drilling or production damage. It is, therefore, clearly of interest to find methods to improve the ratio of success to failure and increase the utilisation of matrix acidising.

Chemistry of Matrix Acidising

It is not the intent of this paper to review the very many complex reactions involved when a heterogeneous rock matrix is attacked or dissolved by acids. Such detailed information can be found in numerous previous publications, many of which are referenced in the extensive listing at the end of this paper. Rather, it is our intention to put the development of matrix acidising in the broader historical context of an increasing knowledge and more sophisticated understanding of the chemical processes involved.

In the early days of acidising, wellsite quality control was almost non-existent and there was little attention paid to such variables as the acid strength or the fact that formations had widely differing mineralogies. Also, there was widespread application of poor practices, like the use of improperly cleaned equipment and rusty tanks⁽¹⁰⁾. Thus, treatments were, indeed, hit-or-miss and the service was little more than a black art. However, as with most technologies, improved understanding came initially from empirical observations at the field level. These were complemented by extensive research and development work carried out by, literally, thousands of scientists and engineers, over many years. Core flow studies, geological and mineralogical investigations, determinations of reaction kinetics, physicochemical modeling of the propagating reaction front, fractal analysis on computer models coupled with solubility testing and reaction product analysis are only some of the many aspects of matrix acidising that have been investigated. Sophisticated, modern day analytical techniques using XRD/XRF, ICP, SEM/EDAX, coupled with computer modeling have allowed detailed examination of the acidising process and provided us with a much better understanding of potential pitfalls and how to avoid them. This knowledge has also helped us design new

acid systems and chemical additives to address factors that may be difficult to control in other ways, thereby improving the success of treatments.

Acid System Design

Carbonate acidising is generally conducted with hydrochloric acid, except, perhaps, in situations where temperatures are very high and corrosion is an issue. In such situations, organic acids like acetic or formic acids are used, since these are much less aggressive than mineral acids. Occasionally, it may also be beneficial to "retard" acid formulations, slowing their reaction rate to allow deeper penetration of live acid or preferential creation of large wormholes through any near-wellbore damage. Various techniques have been employed in an effort to accomplish this including gelling the acid, emulsifying it with oil or using mixtures of acids to take advantage of buffering effects.

The selection of an appropriate acid design for sandstone formations, is a rather more esoteric affair. Part of this problem stems from the complex and heterogeneous nature of most sandstone matrices⁽¹¹⁾. The interactions between the many different mineral species and the injected acid depend not only on the chemical compositions of both but also on temperature, pressure, surface morphology, pore size distribution and pore fluid composition.

Over the years, researchers have run many thousands of core flow tests, solubility tests, and the like, to establish some ground rules on which acids are applicable for which types of mineral. While this is an imperfect science, for the reasons noted above, there is some general consensus on when to use which type of acid. That is not to say that this consensus is always correct. As researchers continue to apply improved test protocols to better replicate actual downhole conditions, some old-established ideas have been challenged or even discredited. For example, it had been widely accepted that hydrated silica, formed as a result of secondary precipitation, is one of the major causes of damage in those acid treatments that fail to produce benefit. More recent work, however, has suggested that partial dissolution of aluminium-rich layers in clay minerals may weaken their structure, resulting in disintegration and generation of mobile fines and it is these that have greater responsibility for the damage.⁽¹²⁾

Over the years, many different acidising systems have been developed for specific applications. In general, the three principal drivers for these developments have been 1) the desire to retard the acid/mineral reactions, thereby achieving greater penetration, and 2) the desire to make the acid less aggressive to well architecture ie. tubulars, wellheads, screens, etc. and 3) the desire to avoid undesirable reactions that could result in formation damage.

Some of the approaches employed to retard the acid have included the use of buffered-HF⁽¹³⁾ systems or organic systems, the use of fluoroboric⁽¹⁴⁾ acid, and the use of⁽¹⁵⁾ mixtures of esters and fluorides to generate HF in-situ by thermal hydrolysis. Other, more exotic efforts have included the use of hexafluorophosphoric acid or hexafluorotitanic acid. For carbonate acidising, mixtures of esters⁽⁶⁵⁾ with enzymes have been used to generate organic acids, in situ. From a stoichiometric perspective, this is not a particularly efficient

option but it may offer some advantages in terms of environmental impact and corrosion.

In general, those systems that generate acid in situ, or that use organic acid blends, also address the problem of corrosion, the second driver, noted above. However, it is worth noting that such systems may still cause corrosion problems on flowback, if they contain no inhibitors.

The third issue, that of mitigating undesirable reactions or their by-products, has spawned many proprietary formulations, as well as changing some of the application guidelines used in matrix acidising. The old “generic” acids consist of mixtures of hydrochloric and hydrofluoric acid, known in the industry as “mud acid”. Traditionally, the ratio of HCl to HF was 4:1 (ie. 12:3 or 6:1 ½). However, Gdanski⁽⁷⁾ and others⁽³⁴⁾ have suggested that, for such conventional formulations, it may be necessary to increase this ratio to as much as 9:1 (ie. 9:1 or 13½:1½). The rationale for these relatively high ratios of HCl:HF is the fact that dissolution of clays by HF mixtures produces many secondary reaction products that can, in turn, re-precipitate in the formation and cause damage. These damaging reaction products, produced by conventional mud acids, are slightly more soluble if the pH is kept low throughout the treatment, and during flowback. One could justifiably ask why we would even consider injecting such an incredible volume of HCl that will not react with the majority of near-wellbore damage or the bulk of a sandstone formation. Such systems can only exacerbate the risk of corrosion and may compromise well integrity by reducing the useful life of well tubulars and pressure control equipment. Fortunately, new acid systems^(8, 9, 17, 18, 19) have been developed that eliminate these problems and excess HCl is no longer required for the sole purpose of reducing secondary precipitates. These are referred to in more detail, below.

Interestingly, formations with high levels of chlorite, an iron-bearing clay, respond best to acid formulations containing no hydrochloric acid whatsoever, either in the preflush or in the main stage of the treatment. Instead, acetic acid rather than hydrochloric acid is recommended, since the former limits iron liberation and thereby reduces the risk of precipitates from iron reaction products. However, Gdanski has recommended that, where acetic acid is used, ammonium chloride needs to be added to the acid to ensure adequate inhibition of any water sensitive clays. More recently, researchers have found that HF formulations containing a blend of phosphonic and citric acids are beneficial in acidising formations containing zeolites, a particularly troublesome aluminosilicate mineral family with strong ion-exchange tendencies. Such systems are members of a family of improved acid systems, specifically designed for use in sandstone reservoirs and reported by numerous authors elsewhere in the literature. They have some special properties that minimize or eliminate many of the problems commonly associated with sandstone acidizing. These acids are retarded, and minimize the reprecipitation of secondary reaction products that normally result from the reaction of HF with clays. They are also much less corrosive and safer to handle, with the pH of the live acid typically in the range of pH 3.5 – 4.5. Therefore, in terms of corrosion inhibitor loading, these new acids requires less inhibitor to protect both coiled tubing

and well tubulars. Due to their lower inherent corrosivity, they are also a much safer option to use in the clean-up of gravel packed wells, since there is a greatly reduced risk of damage to screens.^(8, 9, 20)

One new system combines a powerful iron complexing agent and a scale inhibitor to eliminate the risk of iron precipitation, a common cause of treatment failure. Results obtained with this new system in diverse and very challenging mineralogies were extremely impressive and this has been borne out in the field⁽²¹⁾. Other new systems, that have yet to be field tested, feature biodegradable blends with very low corrosivity. This helps minimise inhibitor loadings, thereby making the acids as environmentally-friendly as possible, fulfilling one of our commitments to improved stewardship of the biosphere. These acids are also safer to handle, reducing risks to personnel and equipment.

Another approach that has merit is the use of non-acid systems for matrix stimulation. These are generally based on chelating agents like EDTA or, more recently, the hydroxyaminocarboxylates^(22, 23). The latter are promoted for their better biodegradability, amongst other things. All of these agents are generally employed to remove calcium-based scales, or a portion of formation mineralogy. However, they are incapable of dissolving clays and other silicates. The types of acid used in matrix acidising are reasonable familiar to many engineers. However there is also a vast collection of chemicals that are routinely added to most acidising systems. The types of additive that are used belong to several principal categories and these are discussed individually, below. Also, Table 1 provides a list of the typical concentration ranges of these same materials.

Corrosion Inhibitors

In general, acids attack steel to produce solutions of (mainly) iron salts while generating hydrogen gas. Depending on the steel metallurgy, type of acid (mineral or organic), its strength and the temperature, the reaction may be more or less vigorous. However, particularly with mineral acids, this attack can lead to the removal of a substantial amount of metal mass, potentially weakening or shortening the lifespan of well tubulars.

In general, the industry has adopted reasonably standard levels of acceptable weight loss due to corrosion, during the specified exposure time. Typical limits are <0.05 lbm/ft² for conventional tubulars (including high alloy steels) and <0.02 lbm/ft² for coiled tubing, to take account of the latter's reduced wall thickness and its different work profile. However, the problem is exacerbated by the fact that the corrosion may not be uniform but may take place in localised “corrosion cells” on the steel surface, or inside microcracks, to produce deep pits. These pits can develop quickly under certain circumstances and may result in pipe perforation and loss of pressure integrity or mechanical strength.

It was the discovery of an effective corrosion inhibitor that sparked the widespread application of acidising in the 1930's⁽³⁾. That first inhibitor was based on arsenic and, while efficient, its use was discontinued for obvious concerns about toxicity and the environment. Arsenic is also a well-known catalyst poison and its presence in produced well fluids caused

concern for refineries. A variety of organic inhibitors has superseded these early arsenic compounds. The majority of these are based on so-called acetylenic alcohols, like octynol and propargyl alcohol, highly reactive molecules containing a carbon-carbon triple bond. The ability of these materials to protect steel is dependent on the proper dispersion of the hydrophobic alcohol in the acid, since it is widely accepted that the protection mechanism involves the creation of an inhibitory film on the metal surface. A number of other materials can further enhance this protection or extend it to fairly high temperatures. These are used in concert with the inhibitors and include certain reducing agents like formic acid and iodides as well as compounds of antimony, an element in the same Periodic Table Group as Arsenic and Phosphorus.

Recently, for certain applications, conventional oilfield, carbon steel has been replaced by more exotic metallurgy, including chrome or even nickel/chrome alloys. Such metals are more resistant to attack by subsurface materials like CO_2 and H_2S but they can be more vulnerable to mineral acids, in particular, and may not be adequately protected by conventional inhibitor packages. Also, environmental legislation has forced a re-evaluation of many corrosion inhibitors and this has meant some trade-off between toxicity⁽²⁹⁾ and overall performance or efficacy. Alone, and in combination, these developments have made it much more difficult to provide required levels of corrosion protection, particularly at high temperatures. For that reason, there is some suggestion that the most effective way to acidise in high temperature wells, in general, is to use organic acids, like formic and acetic, rather than mineral acids. These are much easier to inhibit and are relatively cost-competitive and, at the same time, are biodegradable.

Iron Control Agents

Iron, which is ubiquitous in any oilfield operation, is perhaps the most underestimated problem of all in acidising. Surface tankage and lines, mixing and pumping equipment and well tubulars are all routinely made of steel, which dissolves to a greater or lesser extent in acid. Even considering only the iron that would enter solution as a result of allowable corrosion on clean steel, this may amount to many thousands of mg/litre. In general, clean steel dissolves to produce ferrous (Fe^{2+}) ions, but these can be converted to the much more problematic ferric iron (Fe^{3+}) by dissolved oxygen. Also, surface steel is universally coated with a patina of rust, or ferric oxide, which quickly dissolves to produce ferric ions in acid. In many wells, the tubulars are covered in scale, often of iron compounds, including iron sulphides (in sour wells) and iron carbonates and oxides (in injector wells). Finally, many formations contain iron-bearing minerals, like siderite, ankerite (iron carbonates) or chlorite and these can all be dissolved when contacted by acid. Not to be overlooked, either, is the raw acid itself, which should be clear and water-white but is often delivered already discoloured a greenish-yellow by iron compounds. All of these sources can add up to a significant concentration of iron in the live or partially spent acid.

Unfortunately, several iron reaction products can precipitate from acid as it spends and the pH rises. The most likely is

ferric hydroxide, which forms a gelatinous, plugging precipitate when the acid pH rises about pH 2.2^(19, 27). In sour wells, iron sulphide (FeS) and even colloidal sulphur can precipitate from acid when the pH is in the range of pH 1.9^(24, 25, 26, 28). Much less likely to form is ferrous hydroxide, which only precipitates at close to pH 7, a value unlikely to be encountered even with completely spent acid.

Various chemical methods have been employed to address this issue of iron precipitation. The most common ones include chelation/sequestration and reduction. Typical examples of the former include citric acid, EDTA and its salts, and NTA (Nitrilotriacetic Acid). The reducing agents, on the other hand, include erythorbic and ascorbic acids. For sour wells, various proprietary agents based on thio-acid, carbonyl compounds and tin compounds have been successfully used. Sometimes, combinations of these different agents are used to provide necessary levels of iron control while minimising any risk of unwanted byproducts. In certain proprietary systems, the presence of one or other phosphonic acids, provides additional protection against iron precipitation. These agents form water soluble complexes with iron and they also operate at sub-stoichiometric levels to inhibit the nucleation and growth of potential precipitates.

Clay Stabilizers

As noted previously, sandstone formations are particularly heterogeneous, generally with quartz as the primary skeletal mineral and various carbonates, clays, or feldspars acting to cement the sand grains together. In addition, clay minerals may be suspended in the pore filling fluid or may line the pore spaces. Particularly in the latter case, these clays can react very badly when contacted by injected fluids, generally due to ion exchange or partial dissolution. Often this results in the dis-aggregation, disintegration or swelling of the clay and can plug the pore spaces and pore throats. For this reason, many acidising formulations contain clay stabilisers to mitigate the problem. The simplest clay stabilisers are salts like ammonium and potassium chloride, although the latter cannot be used in situations where HF is used due to the risk of secondary precipitates, like potassium fluorosilicate. In fact, it is much more common to incorporate into the treating fluids any one of several synthetic materials that have been found to prevent clay swelling. These materials are usually cationic, in nature, like quaternary amines or polymers with similar active groups.

Other materials have been used to try to prevent the migration of mobile clays or silica fines through the matrix since these can cause problems if they accumulate in the near wellbore. One such approach, in use for many years, has been to modify the acidising system itself by replacing simple hydrofluoric acid with fluoroboric acid (HBF_4). The latter is slowly hydrolysed to HF, reducing the reaction rate of HF and apparently helping to fuse clay platelets together⁽³²⁾. Recent published work, however, suggests that fluoroboric acid can produce several adverse reactions with other formation minerals, forming secondary precipitates⁽¹⁹⁾. Organosilane⁽¹⁴⁾ materials, that were discovered over 15 years ago, are being used to accomplish, essentially, the same fines-fixing effect.⁽⁵⁵⁾

Surfactants

The term “surfactant”, a contraction of Surface-Active Agent, encompasses a very diverse group of chemicals. These include foaming agents, water-wetting agents, oil-wetting agents, emulsifiers, non-emulsifiers (or demulsifiers) and anti-sludge agents, to name a few. All of these agents have effects on surface and/or interfacial tension. For example, water-wetting surfactants may lower the surface tension of aqueous fluids from around 70 dyne/cm to values of around 25 dyne/cm, thereby improving the ability of the treating fluid to penetrate small pores and to react with the matrix constituents. Such surfactants also improve the recovery of these same fluids after the treatment. This is particularly important in lower pressure wells, which may have difficulty in driving aqueous fluids from the pore spaces, resulting in a so-called “water block”. For these reasons, it is widely accepted practice to include water-wetting surfactants in almost all matrix acid treatments. In many cases, these surfactants have replaced volatile solvents like methanol that were used in the past to enhance clean-up of gas wells, partly by virtue of their forming low boiling point azeotropic mixtures.

Oil wetting surfactants, in contrast, have been used in some situations to retard the reaction of acid with carbonate rock. A similar effect is obtained when emulsifying surfactants are used to generate oil-out emulsions, where acid is emulsified as the disperse phase in a hydrocarbon continuous phase. Such systems are used primarily in acid fracturing where it is important to retard acid spending in order to achieve etching along as much of the created fracture length as possible. Such agents are generally not used in matrix acid treatments.

Demulsifiers and non-emulsifiers are all designed to prevent, or facilitate the break of, the emulsions that tend to form between crude oil and live, or spent, acid fluids. Such emulsions can be very viscous, even quasi-solid, and may plug the pores of the treated matrix. API test methods are used to allow comparison of the efficacy of various blends of demulsifying surfactants. The optimal additive can often achieve an almost instantaneous separation of the oil and water phases with a clean, well-demarcated interface. Occasionally, however, particularly with heavy, asphaltenic crudes, certain components of the oil are precipitated out, generally forming a sludge at the oil:acid interface. In such cases, anti-sludge agents are useful since these prevent the formation of sludge or help redissolve it. In severe cases, where iron is present in the acid formulations, organic acids may be required or a low pH active iron sequestering agent, in combination with the surfactant, must be used.

Foaming agents are widely used in acidising treatments, often to provide diversion. In such cases, the foam is typically 60-75 quality and may be pumped in discrete diversion stages to provide better treatment distribution.

Diverting Agents

As just noted, it is not always possible to direct the acid treatment to the intervals that require it most, particularly in a long interval. Differential permeability of the numerous zones, or even sub-layers within the same zone, will have a tendency to predispose to a certain injection profile. Of course, the more

permeable intervals are those that will tend to take most of the fluid and, as acid removes any damage, these same intervals will become even more permeable, leading to a snowball effect. At the limit, the intervals with greater damage, or lower permeability may end up receiving none of the acid. Ideally, we would like to re-distribute the acid injection profile in such a way that the interval is uniformly treated. Of course, there are several things to consider. One could argue that it is the higher permeable intervals that are likely to have been most damaged during the drilling process. These same intervals will also benefit the most from the injected acid since they have higher intrinsic productive capacity. On the other hand, diverting acid into tighter sections may not produce the desired results since these intervals may never produce at economic rates, even with no damage or moderate, true stimulation.

As with most things in life, a balance needs to be struck. This is usually accomplished by allowing a certain volume of acid to flow into and thereby remediate the higher permeability intervals and subsequently diverting the flow to lower permeability zones. The agents used to accomplish this are numerous. The most common materials are particulates, usually insoluble in acid but soluble in hydrocarbon for easy clean-up. Such agents include benzoic acid, naphthalene, oil-soluble resin, gilsonite and wax beads.⁽¹³⁾ Other systems have included polymers that cross-link as pH or calcium ion level rises or viscoelastic surfactant systems that have similar characteristics.^(40,43) For injector wells, organic acid salts may be used, since these are insoluble in acid but readily dissolve in injection water and can be flushed away⁽⁶⁷⁾. Occasionally, for perforated completions, ball sealers are used, although the length of the perforated interval and the practical injection rate may limit their use⁽⁵⁴⁾. The other widely used diverting agent, already mentioned, is foam, usually at 60-75 quality.⁽⁵⁵⁾

There are other methods that can help re-distribute an acid treatment. One is simply to pump at the maximum possible rate at the maximum possible pressure, below fracturing pressure, throughout the treatment. Thus, very low rates may be used at the start of a treatment but, as injectivity improves in response to acid, the rate is increased to maintain the pressure at the prescribed value^(46, 49). Another technique is to use some form of selective isolation tool such as a straddle packer, or even sets of opposing cups, to try to bracket the interval that is exposed to the acidising fluid. More recently, several authors have identified the benefits of jetting techniques to improve the placement selectivity of stimulation fluids, particularly in gravel packed completions.^(20, 30)

Miscellaneous Additives

There are several additional additives that may, occasionally, be required in acidising formulations. These include mutual solvents, which are often used with problematic crudes that are prone to forming emulsions. Mutual solvents are usually based on alcohols or ethers and are intermediate, in terms of polar character, between water and oil. Thus, they act as good co-solvents and are miscible with both phases, often dramatically reducing emulsion viscosities and improving the performance of demulsifiers.

Sometimes, particularly with heavier asphaltic crudes, it is necessary to use aromatic solvents, usually in concert with appropriate surfactants, to minimise the risk of emulsions and sludges. Historically, the solvents of choice have been xylene or toluene since these are cheap and readily available. However, increased concerns over the toxicity of these materials has led to the introduction of environmentally-friendly solvents based on terpenes. The latter are natural, plant-derived hydrocarbons with exceptional solvency and they have the benefit of being readily biodegradable.⁽⁵⁸⁾

In matrix acidising, friction reducers are generally only used in coiled tubing operations, or in very high rate bullhead treatments. However, these agents can certainly be beneficial under such circumstances. Various studies have shown friction reduction of 70-80 percent with minimal concentrations of friction reducer. This translates to higher injection rates without exceeding surface pressure limitations and is generally regarded as beneficial in matrix acid treatments, as noted elsewhere.

Software Developments

As should be clear from the discussion above, there are many rules that must be learned by an individual who wishes to become adept in matrix acidising. There may be only a relatively small number of variants on the theme of acid type and strength, but there is a bewildering array of additives that can, should, or sometimes must, be included in an acidising formulation.

In each category, there are many individual additives that can be selected on the basis of well conditions and the principal fluid composition. The rules that govern the use of these additives are complex but, generally, factors that are considered include:

- a) Type of acid
- b) Acid strength
- c) Formation temperature
- d) Mineralogy
- e) Permeability
- f) Rock Strength (degree of consolidation)
- g) Formation pressure
- h) Formation fluid composition
- i) Presence/absence of scale
- j) Type of completion
- k) Completion metallurgy
- l) Treatment conduit (eg. tubing, coiled tubing)
- m) Additive compatibility

Many of these same factors, individually or in combination, guide the selection of an appropriate concentration at which the specific additives will be used.

With so many variables, and so many additives to choose from, the number of possible combinations is enormous. There may, of course, be more than one appropriate design for a certain set of conditions. However, there are also many totally inappropriate designs that could be arrived at by individuals who do not know all of the rules, or have insufficient experience. Application of one of these bad designs could

easily result in one of those “disastrous” acid jobs, mentioned previously.

Given that acidising represents one of the most effective methods for production enhancement, most major service companies and some major operators have developed software tools to help in the design of acid treatments. The design process represents no more than the application of a series of logical decisions, based on particular rules, so it lends itself to computer assistance. Admittedly, the number of rules is quite large but, once built, a computer program that encompasses these rules is capable of providing excellent acid designs. There are numerous benefits to applying this approach, including standardisation, improved training and assured competence.⁽²¹⁾

Other Developments

While the implementation of an appropriate and specific design methodology can do much to improve results, it cannot achieve success alone. To a large extent, it is the field implementation of any technology that dictates whether or not it is successful.

Unfortunately, field practices often lag behind what would be considered normal standards of quality control when dealing with the multi-million dollar asset that an oil- or gas well represents. Most experienced employees, both of service and operating companies, who have been involved in operations, could probably attest to routine application of less-than-perfect field practices. Use of rusty tanks to mix acid, no filtration of injection fluids, no pickling of pipe and surface lines, no monitoring of the well's response to the treatment – these are only examples of some of the shortcomings that can nullify the potential benefit of performing an acid treatment, in the first place. Thus, the transfer of technology and the training, not only of engineers but of field personnel also, are paramount. As with any oilfield operation, attention to detail and commitment to quality, all the way from corporate office to wellhead, are key components to success.⁽²¹⁾

One exciting development of the newer acid systems is the ability to prepare them “on-the-fly” from relatively small quantities of concentrated liquid additives⁽⁹⁾. This can be done using a customised blending/proportioning unit to produce the final acid. The unit requires only a supply of fresh water, which typically constitutes from 70-90 percent of the final acid treating solution, depending upon the required blend. In the case of sandstone acidising, no HF mixtures need to be prepared in mix tanks on board the rig/platform, or at the wellsite, and treatments can be accurately sized in accordance with specific well requirements and reservoir response. Furthermore, problems of disposal of unused acid, buyback/restocking charges, not to mention the hazards associated with the cutting and blending of dangerous chemicals, like ammonium bifluoride and concentrated HCl, are eliminated. The safety advantages, as well as the logistics and environmental benefits, of using this approach are convincing enough in themselves.^(21,58)

Measuring Success

As noted previously, the success of matrix acidising depends, to a great extent, on the combination of operational excellence

and management commitment. However, it is still the exception rather than the rule, for acid treatments to be monitored in real time to gauge their ability, and efficiency, in removing skin damage. Paccaloni⁽¹⁵⁾ proposed a methodology many years ago to accomplish this, building on earlier work by McLeod and Coulter⁽⁵¹⁾, and, while not perfect, it certainly represents one possible means to assess the efficacy of an acid treatment, in real time. An alternative technique was suggested by Prouvost and Economides⁽¹⁶⁾ and several more recent publications have proposed similar strategies.

Allowing for certain assumptions, these techniques can certainly be used, particularly with the ready availability today of wellsite monitoring equipment, data acquisition units and computers. These permit the estimation, at least, of real-time skin evolution, as the treatment proceeds using only pressure and rate measurements. The interpretation or analysis is, of course, complicated by the use of multiple stages, foam fluids, particulate diverters and the like, but strategies for handling such effects are available and service company software is capable of modeling and accounting for these changes. The beauty of all such strategies is the ability to stop the treatment when the desired result, in terms of computed skin reduction has been achieved. When coupled with the ability, noted previously, to prepare acid formulations on-the-fly, without the need for large volumes of pre-mixed acid in surface tankage, this represents a particularly elegant method to optimise treatments.

Of course, ultimately, the only way to evaluate the success or failure of an acid treatment is to compare stabilised production (or injection) rates, before and after the treatment. Particular care should be taken to ensure that true comparisons are made and that well conditions are carefully noted. The success of a stimulation treatment may manifest itself in many ways and it is important that both operator and service provider are privy to results so that future improvements can be made, if necessary.

Closing Comments

Given the progress in understanding the chemistry and physics of matrix acidising and the many developments in design software and wellsite quality control, one may justifiably ask why acidising is not applied on a much more routine basis. Much of the blame can be attributed to a poor understanding of the potential benefits and relatively low cost of acidising. However, probably the most persistent problem is related to the various “horror stories” of dramatically unsuccessful acid treatments that tend to circulate in the industry, long after the event. In such situations, the exact well details and circumstances of the treatment are often forgotten but the memory of the failure persists. In examining the reasons for the disparity, between the successful application of matrix acidising and those treatments that were unsuccessful, most authors have arrived at similar conclusions. The principal reasons for poor response following acidising are:

- a) Poor candidate selection
- b) Lack of mineralogical information
- c) Wrong acid design (strength, volume, etc)
- d) Use of inappropriate acid additives

- e) Insufficient iron control
- f) Use of contaminated/dirty fluids or neglecting to pickle tubing string
- g) Improper placement of acid (eg. lack of diversion, plugged perforations)
- h) Long shut-in time without recovering injected fluids

While there are several reasons, above, related to factors like insufficient information or poor field procedures, the majority relates to engineering and treatment design issues. Clearly, selection of an inappropriate candidate well for acidising will, at best, produce mediocre or disappointing results, at worst, disastrous ones. Thus, it is first important to examine actual well performance (based on DST, production tests or similar) and compare this with expectation, based on theoretical performance. In older wells, production history should be examined, looking, in particular, for anomalous behaviour or events superimposed on the general trend.

Engineering staff can easily undertake such exercises using standard well system analysis software. The same software can also identify wells where stimulation would prove ineffective due to completion limitations (inadequate perforations, tubing size, chokes, etc). This is not “rocket science” and it requires relatively little effort to identify an under-performing well. The next step is to identify the reason for that under-performance and establish whether it can be remediated economically by acidising. It bears repeating, that the best stimulation candidates are the good wells – the high producers that are not producing at optimal rates due to some sort of formation damage. Therefore, wells with high formation permeability and substantial reservoir pressure, but with high skin, are the wells to target initially, in any stimulation campaign.

Unfortunately, because they are perceived to represent the highest risk in the event of failure, these same wells are rarely the ones that are offered up for acidising. Instead, the low permeability, low pressure, old wells are usually the proffered candidates – the wells with the highest likelihood of failure and the least upside productive potential in the event of success.

Matrix acidising is the most cost-effective way to enhance oil production from damaged wells, of that, there is little doubt. Admittedly, today, it is not yet a perfect discipline but we should never let a bad result from one well dissuade us from pursuing a campaign of production enhancement. Rather, we should integrate the production gains from a multi-well program to truly assess the benefits achievable with this sadly, under-utilised technique.

In any endeavour, there is an element of risk and matrix acidising is no different, in that regard. It is always important to understand the risks and take appropriate measures to minimise them. However, because of some of the factors mentioned above, it is still possible to have unexpected results. Such events should be seen in the broader context as opportunities to learn something for future treatments.

In an industry like ours, where there will always be unknown variables, there will always be treatment failures. However, as in the past, we must always push the envelope to expand our

knowledge. If we never have failures, it could be argued that we are probably not pushing the envelope far enough.

Conclusions

- 1) Matrix acidising, with the appropriate systems in correctly identified candidate wells, is the most cost-effective way to enhance oil production in both sandstone and carbonate reservoirs.
- 2) Tremendous progress has been made in understanding the chemistry and the physics of the acidising process and this, coupled with improvements in wellsite implementation, has translated into better success.
- 3) The ability of a low-cost, well-designed acid treatment to spectacularly improve well productivity is still poorly appreciated by many people.
- 4) The use of computer software that encompasses all the known rules and guidelines for sandstone acidising can greatly improve the success ratio by eliminating inappropriate designs and standardising treatments.
- 5) New acid systems with much improved performance have been developed specifically to address many of the problems inherent in sandstone acidising. These new acids help eliminate many of the problems of conventional acids.
- 6) This paper reviews the many changes that have taken place in acidising since its first application and provides an overview of the additives and systems used today.

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References:

1. Frasc H. 'Increasing The Flow of Oil Wells' US Pat 556, 669 (March 17, 1896)
2. Williams, B.B., Gidley, J.L., Schechter, R.S. 'Acidizing Fundamentals' Monograph Volume 6, chapters 3, 4, 8 and 9, SPE 1979
3. Kalfayan L, 'Production Enhancement With Acid Stimulation' Pennwell Corporation (2000) 51
4. A. Daneshy, 'Economics of Damage Removal and Production Enhancement' paper SPE 30234, presented at the Society of Petroleum Engineers European Formation Damage Symposium, The Hague, Netherlands, May 15-16, 1995
5. Smith C.E. and Hendrickson AR 'Hydrofluoric Acid Stimulation of Sandstone Reservoirs' JPT (Feb 1965) 215
6. Crowe C.W., 'Precipitation of Hydrated Silica from Spent Hydrofluoric Acid: How Much of a Problem Is It?' JPT (Nov 1986) 1234
7. Gdanski R.D., 'Fluosilicate Solubilities Impact HF Acid Compositions' SPE 27404, Symposium on Formation Damage Lafayette Feb 7-10 (1994)
8. Di Lullo, G. and Rae, P. 'A New Acid for True Stimulation of Sandstone Reservoirs,' paper SPE 37015 presented at the 1996 SPE Annual Technical Conference and Exhibition held in Denver, CO, 6-9 October
9. Ross, D. and Di Lullo, G. 'HV:HF' Acid Treatments, Proven Successful in South America,' paper IBP-SE-076/98 presented at the 1998 Rio Oil & Gas Conference held in Rio de Janeiro, Brazil, 5-6 October
10. Ely J., Stimulation Treatment Handbook, PennWell Books (1987)
11. Ali Syed A., 'Sandstone Diagenesis, Applications to Hydrocarbons Exploration and Production', USA (1981)
12. Gdanski R.D. and Shuchart C.E.: 'Advanced Sandstone-Acidizing Designs With Improved Radial Models' SPE 52397, San Antonio, Texas 5-9 October 1997
13. Doerler, N and Prouvost L Diverting Agents - Laboratory Study and Modeling of Resultant Zone Injectivities SPE 16250.
14. Kalfayan L.J. and Watleins D.R.: 'A New Method for Stabilizing Fines and Controlling Dissolution During Sandstone Acidizing', SPE 20076 (1990)
15. Paccaloni G.: 'New Method Proves Value of Stimulation Planning', OAJ, Nov 1979 155-60
16. Prouvost L and Economides M.J.: 'Real-Time Evaluation of Matrix Acidising Treatments Petroleum Science and Engineering, Nov 1987
17. Kume, N., van Melsen, R., Erhahon, L., and Abiodun, A.F. 'New HF Acid System Improves Sandstone Acidizing Success Ratio by More than 400% Over Conventional Mud Acid System in Niger Delta Basin,' paper SPE 56527 presented at the 1999 SPE Annual Technical Conference and Exhibition held in Houston, TX, 3-6 October
18. Martin A.N. and Smith K.L., 'New HF Acid System Produces Significant Benefits in Nigerian Sandstones,' SPE Paper 38595, presented at SPE Annual Technical Conference & Exhibition San Antonio, Texas 5-8 October 1997
19. Nasr-El-Din H.A., Al-Dahlan M.N., As-Sadlan A.M., Al-Zamil H.A., 'Iron Precipitation During Acid Treatments Using HF-Based Acids presented at SPE International Symposium and Exhibition, Lafayette, February 20-21, 2002
20. Stanley F.O., Troncoso J.C., Martin A.N., Omar Ali Jamil, 'An Economic, Field-Proven Method For Removing Fines Damage From Gravel Packs' paper SPE 58790, presented at the SPE International Symposium, Lafayette, February 23-24, 2000
21. Rae P. and Di Lullo G.: Achieving 100 Percent Success In Acid Stimulation Of Sandstone Reservoirs, SPE 77808, Melbourne, Australia, 8-10 October 2002
22. Frenier W.W., Fredd C.N., Chang F.: 'Hydroxyaminocarboxylic Acids Produce Superior Formulations for Matrix Stimulation of Carbonates at High Temperatures' SPE 71696, New Orleans, Louisiana 30 September-3 October 2001
23. Frenier W. W., Wilson D., Crump D. and Jones L.: 'Use of Highly Acid-Soluble Chelating Agents in Well Stimulation Services' SPE 63242, Dallas, Texas 1-4 October 2000
24. Delorey J.R., Vician D.N., Metcalf A.S.: 'Acid Stimulation of Sour Wells' SPE 75697, Canada, Alberta, Calgary, 30 April-2 May 2002
25. Nasr-El-Din H.A., Al-Mutairi S.H., Al-Driweesh S.M.: 'Lessons Learned from Acid Pickle Treatments of Deep/Sour Gas Wells' SPE 73706, Lafayette, Louisiana 20-21 February 2002
26. Salma T.: 'Cost Effective Removal of Iron Sulfide and Hydrogen Sulfide from Water Using Acrolein' SPE 59708, Midland, Texas 21-23 March 2000
27. Crowe C.W., 'Evaluation of Agents for Preventing Precipitation of Ferric Hydroxide from Spent Treating Acid' JPT (April 1985) 691-695
28. Brezinski M. M.: 'Chelating Agents in Sour Well Acidizing: Methodology or Mythology' SPE 54721, The Hague, The Netherlands, 31 May-1 June 1999
29. Brezinski M.M.: 'New Environmental Options for Corrosion Inhibitor Intensifiers' SPE 52707, Austin, Texas, 28 February - 3 March 1999

30. Mascarà S., D'Ambrosio Zambelli A., A., Gili V., Loving S., Dossena M.: 'Acidizing Deep Open-Hole Horizontal Wells: A Case History on Selective Stimulation and Coil Tubing Deployed Jetting System' SPE 54738, The Hague, The Netherlands 31 May-1 June 1999
31. Mitchell W.P., Martin A.N. and Stemberger D.J.: 'Is Acid Placement Through Coiled Tubing Better Than Bullheading?', IcoTa, Houston, March 2003
32. Thomas R L and Crowe C W, 'Matrix Treatment Employs New Acid System for Stimulation and Control of Fines Migration in Sandstone Formations' JPT 1981 (1491 - 1500)
33. Thomas R.L., Nasr-El-Din H.A., Lynn J.D. and Mehta S.: 'Precipitation During the Acidizing of a HT/HP Illitic Sandstone Reservoir in Eastern Saudi Arabia: A Laboratory Study' SPE 71690, New Orleans, Louisiana 30 September-3 October 2001
34. Thomas R.L., Nasr-El-Din H.A., Mehta S., Hilab V. and Lynn J.D.: 'The Impact of HCl to HF Ratio on Hydrated Silica Formation During the Acidizing of a High Temperature Sandstone Gas Reservoir in Saudi Arabia' SPE 77370, San Antonio, Texas, 29 September-2 October 2002
35. Nasr-El-Din H.A., Al-Mulhem A.A. and Lynn J.D.: 'Evaluation of Clay Stabilizers for a Sandstone Field in Central Saudi Arabia' SPE 39584, Lafayette, Louisiana, 18-19 February 1998
36. Hashem M.K., Nasr-El-Din H.A., Hopkins J.A.: 'An Experience in Acidizing Sandstone Reservoirs: A Scientific Approach' SPE 56528, Houston, Texas, 3-6 October 1999
37. Nasr-El-Din H.A., Lynn J.D., Hashem M.K., Bitar G.E., Al-Ali A.A.: 'Lessons Learned from Descaling Wells in a Sandstone Reservoir in Saudi Arabia' SPE 74690, United Kingdom, Aberdeen, 30-31 January 2002
38. Wang X. G., Zou H. L., Checg X. S., Li S. Z., Zhang B. P., Hill A.D., Zhu D.: 'A New Approach for Matrix Acidizing of Water Injectors in Low-permeability Sandstone Fields' SPE 73780, Lafayette, Louisiana 20-21 February 2002
39. McLeod H.O.: 'Matrix Acidizing', *Journal of Petroleum Technology* (Dec. 1984): 2055-69
40. Chang F.F., Acock A.M., Geoghagan A., Huckabee P.T.: 'Experience in Acid Diversion in High Permeability Deep Water Formations Using Visco-Elastic-Surfactant' SPE 71691, New Orleans, Louisiana 30 September-3 October 2001
41. Smith P.S., Cowie L.G., Bourne H.M., Grainger M. and Heath S.M.: 'Field Experience with a Combined Acid Stimulation and Scale Inhibition Treatment' SPE 68312, UK, Aberdeen 30-31 January 2001
42. Nasr-El-Din H.A., Lynn J.D. Taylor K.C.: 'Lab Testing and Field Application of a Large-Scale Acetic Acid-Based Treatment in a Newly Developed Carbonate Reservoir' SPE 65036, Houston, Texas, 13-16 February 2001
43. Chang F., Qu Q. and Frenier W.: 'A Novel Self-Diverting-Acid Developed for Matrix Stimulation of Carbonate Reservoirs' SPE 65033, Houston, Texas, 13-16 February 2001
44. Al-Anazi H.A., Nasr-El-Din H.A., Hashem M.K., Hopkins J.A.: 'Matrix Acidizing of Water Injectors in a Sandstone Field in Saudi Arabia: A Case Study' SPE 62825, Long Beach, California 19-23 June 2000
45. Nasr-El-Din H.A., Fadhel B.A., Al-Humaidan A.Y., Frenier W.W., Hill D.: 'An Experimental Study of Removing Iron Sulfide Scale from Well Tubulars' SPE 60205, Aberdeen, UK, 26-27 January 2000
46. Paccaloni G.: 'A New , Effective Matrix Stimulation Diversion Technique', SPE 24781, Washington, DC, 4-7 October 1992
47. Jones A.T., Davies D.R.: 'Quantifying Acid Placement: The Key to Understanding Damage Removal in Horizontal Wells' SPE 50975, USA, Louisiana 14-15 February
48. Boles J. L. et al. USP 6,443,230, 3rd Sept 2002.
49. Tambini, M.: 'An Effective Matrix Stimulation Technique for Horizontal Wells', SPE 24993, Cannes, France, 16-18 November 1992
50. Andersen K.I.: 'Maximizing Production From Gravel Packed Wells at Gullfaks Field by Stimulation and Installation of Multiphase Booster Pumps' SPE 50682, Hague, The Netherlands, 20-22 October 1998
51. McLeod H.O. and Coulter A.W.: 'The Stimulation Treatment Pressure Record - an Overlooked Formation Evaluation Tool', *Journal of Petroleum Technology* (Aug. 1969): 951-600
52. Hall B.E. and Dill W. R.: 'Iron Control Additives for Limestone and Sandstone Acidizing of Sweet and Sour Wells' SPE 17159, Society of Petroleum Engineers, 1988
53. Stanley F.O., Ali S.A. and Boles J.L.: ' Laboratory and Field Evaluation of Organosilane as a Formation Fines Stabilizer' SPE 29530, presented at the Society of Petroleum Engineers Production Operations Symposium, Oklahoma City, OK, April 1995
54. Bilden D.M. *et al.*: 'New Water-Soluble Perforation Ball Sealers Provide Enhanced Diversion in Well Completions' SPE 49099, prepared for presentation at the Society of Petroleum Engineers Annual Conference and Exhibition, New Orleans, LA, Sept. 27-30, 1998
55. Thompson K. and Gdanski R.D.: 'Laboratory Study Provides Guildelines For Diverting Acid With Foam, *SPE Production & Facilities* (Nov. 1993)
56. Huang T., McElfresh P.M. and Gabrysch A.: 'High Temperature Acidization to Prevent Fines Migration', SPE 73745, Lafayette, Louisiana, 20-21 February 2002
57. Rae P., Di Lullo G and Ahmad A.J.: 'Eliminating Environmental Risks in Well Construction and Workover', SPE 77812, Melbourne, Australia, 8-10 October 2002
58. Rae P., Di Lullo G and Ahmad A.J.: ' Towards Environmentally-Friendly Additives for Well Completion and Stimulation Operations', SPE 68651, Jakarta, Indonesia, 17-19 April 2001
59. Shuchart C.E.: 'Chemical Study of Organic-HF Blends Leads to Improved Fluids', SPE 37281, Houston, Texas, 18-21 February
60. Underdown D.R., Hickey J.J. and Kalra S.K.: 'Acidization of Analcime-Cemented Sandstone, Gulf of Mexico', SPE 20624, New Orleans, LA, 23-26 September 1990
61. Coulter G.R. and Jennings A.R., Jr: 'A Contemporary Approach to Matrix Acidizing', SPE 56279, San Antonio, Texas, 5-8 October
62. Abrams A., Lybarger J.H., Neasham J.W. and Richardson E.A.: 'The Development And Application Of A High pH Acid Stimulation System For A Deep Mississippi Gas Well', SPE 7565, Houston, Texas, 1-3 October
63. Wehunt C.D., Van Arsdale H., Warner J.L. and Ali S.A.: 'Laboratory Acidization Of An Eolian Sandstone At 380 Degrees F', SPE 25211, New Orleans, USA, 2-5 March 1983
64. United States Patent 3,889,753, Richardson, Buffer Regulated Mud Acid
65. Harris R.E. and McKay I.D.: 'New Applications for Enzymes in Oil and Gas Production', SPE 50621, The Hague, The Netherlands, 20-22 October 1998
66. Shuchart C.E. and Gdanski R.D.: 'Improved Success in Acid Stimulations with a New Organic-HF System', SPE 36907, Milan, Italy, 22-24 October
67. Copper R.E. and Bolland J.A.: 'Effective Diversion during Matrix Acidization of Water and Injection Wells OTC 4795, (1984)

Table 1
ACID MATERIALS AND ADDITIVES

Typical Concentration Ranges For Acidising Additives		
Group & Additive Name	Temperature Range, Deg F BHCT	Typical Concentration Range (Guideline Only)
Corrosion Inhibitors		
Inhibitor for Organic Acids	to 400 F	1 - 15 gpt
Inhibitor for Mineral Acids	< 350 F with Inhibitor Aids	1 - 25 gpt
Environmentally Friendly Inhibitor	< 220 F, higher with inhibitor Aids	1 - 20 gpt
High Temperature Inhibitor	400 F	1 - 15 gpt
Sulphide Scavenger	< 250 F or Test	3 - 20 gpt
Inhibitor Aids		
Formic Acid	>200 F	5 - 100 gpt
Potassium Iodide	>200 F	5 - 100 pptg
Copper-Based Inhibitor Aid	>200 F	5 - 50 gpt
Antimony-Based Acid Inhibitor Aid	>200 F	1 - 30 gpt
Iron Control		
Glacial Acetic Acid	< 225 F	10 - 20 gpt
Erythorbic acid	< 350 F	10 - 40 pptg
Citric Acid	< 350 F	10 - 150 pptg
NTA	< 350 F	25 - 350 pptg
EDTANa4	< 350 F	5 - 100 pptg
Clay Stabilizers		
KCl	No limit	2 - 5% BWOW
Ammonium Chloride	No limit	2 - 4% BWOW
KCl Substitute	< 350 F	1 - 2 gpt
Cationic Polymer	< 350 F	0.25 - 2 gpt
Organosilane Fines Stabilizer	< 350 F	1 - 10 gpt
Diversion		
Benzoic Acid Flakes	< 350 F	1 - 2 lbs / gallon
Organic Acid Salt	< 350 F	1 - 2 lbs / gallon
Wax Beads	< 200 F	0.25 - 2 lbs / gallon
Gilsonite	< 330 F	1 - 2.5 lbs / gallon
Ball Sealers	< 400 F	1 / perforation (min)
Biodegradable Ball Sealers	< 300 F	1 / perforation (min)
Foam		75 - 85% Quality
Friction Reducers		
Anionic Friction Reducer	< 300 F	0.125 - 1 gpt
Anti Sludge Agents		
Anionic Aromatic Surfactant	< 300 F	2 - 10 gpt
Foaming Agents		
Anionic Surfactant	< 350 F	1 - 10 gpt
Non-ionic Surfactant	< 350 F	1 - 10 gpt
Cationic Surfactant	< 350 F	1 - 10 gpt
Solvents		
EGMBE	< 250 F	50 - 100 gpt
Terpene Solvents	< 350 F	50 - 250 gpt
Surface Tension Reducers		
Non-ionic Surfactant	< 350 F	0.5 - 10 gpt
Anionic Surfactant	< 350 F	0.5 - 10 gpt
Zwitterionic Surfactant	< 300	1 - 5 gpt

Non-Emulsifiers		
Anionic	< 350 F	1 - 10 gpt
Non-ionic	< 350 F	1 - 10 gpt
Cationic	< 350 F	1 - 10 gpt
Pickling		
Neutral Derusting and Pickling Solution	< 400 F	6 - 12%